Sprint 1 – Project Proposal

Project Title

Student Result Management System for Educational Institutions

Team Members

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Business Context

This application is developed for an educational institution (college or training centre) that needs a reliable digital system to manage student academic results.

Currently, many institutions rely on spreadsheets or manual calculations, which are error-prone and inefficient. This application is developed for an educational institution such as a college or training centre to efficiently manage student academic results. Many institutions currently rely on spreadsheets or manual calculations, which are time-consuming, error-prone, and difficult to maintain as student numbers increase. These methods often lead to data inaccuracies, delays in result processing, and challenges in generating reports.

The proposed system provides a centralized and automated solution for recording, calculating, and managing student results. It improves accuracy, reduces administrative workload, and ensures timely and transparent access to academic information for faculty and student

Application Idea

The Student Result Management System is a CRUD-based web application that allows staff to create, read, update, and delete student academic records.

The system enables administrators to manage student information and marks, automatically calculate results, and retrieve data efficiently through a web interface. The Student Result Management System is a CRUD-based web application designed to manage student academic records in a structured and efficient manner. The system allows authorized staff to create, read, update, and delete student information, subject details, and marks through a secure web interface.

Administrators can manage student profiles, enter and modify marks, and automatically calculate totals, grades, and pass/fail status based on predefined evaluation rules. The system also enables quick retrieval of student results and generation of reports, reducing manual effort and minimizing calculation errors.

By centralizing academic data, the application improves data accuracy, accessibility, and consistency, while providing a user-friendly platform for managing student results effectively.

Purpose of the Application

The purpose of this application is to:

- Improve efficiency in managing student results
- Reduce manual errors in calculations
- Provide fast access to accurate academic data
- Support staff with a structured, database-driven solution

This application represents a strong use case for a CRUD system, as student records must be continuously created, updated, viewed, and removed.

Business Questions

The application is designed to help the business answer the following questions:

1. How can student results be stored securely and consistently?
2. How can result calculation errors be reduced?
3. How can staff quickly retrieve a student's academic performance?
4. How can academic data be updated efficiently when corrections are needed?
5. How can historical student records be maintained digitally?

Target Users (Personas)

Persona 1: Academic Administrator

- **Name**: Reilly Williams
- **Role**: College Administrator

- **Goals**:

➓ Enter and manage student results efficiently

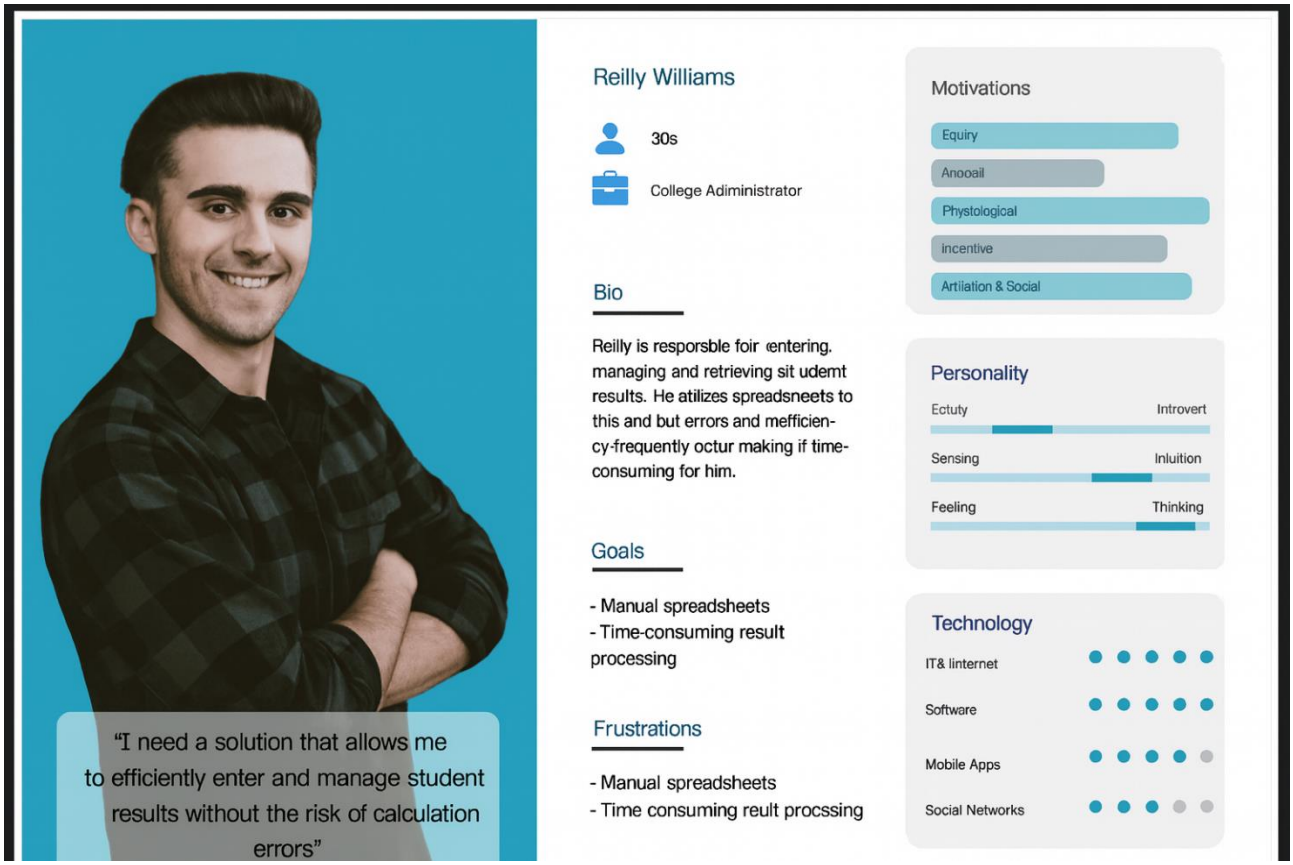
➓ Avoid calculation errors

➓ Quickly retrieve student records

- **Frustrations**:

➓ Manual spreadsheets

➓ Time-consuming result processing



Persona 2: Course Tutor

- **Name**: John Patel
- **Role**: Lecturer / Tutor
- **Goals**:

❿ View student performance

❿ Check pass/fail status

❿ Identify students needing support

- **Frustrations**:

John Patel

40s

Lecturer / Tutor

Bio

John teaches multiple courses and needs to review results for students in his classes. He helps students in need of easis-tance and prefers a stheamine way to access their performence.

Goals

- View student performance
- Check pass/tail status
- Identrty students heeding support

Frustrations

- Delayed access to results
- inconsistent record formats

"I need to easily access

Motivations

Equity

Antead

Phydotogica!

Incentive

Personality

Extrovert Introvert

Check pass/fail status

Identify students needing sup-

Personality

Extrovert

sensing

Manialty Atiow

persona 3: Williams john

Course mediator:

Bio

Reilly oversees the entry, verification, and retrieval of student academic results. He relies heavily on spreadsheets, which often leads to errors, duplicated work, and long processing times. His role requires precision, but manual systems slow him down.

Goals

- Reduce manual data entry
- Improve accuracy in result processing
- Access student records quickly.

Frustrations

Frequent spreadsheet errors

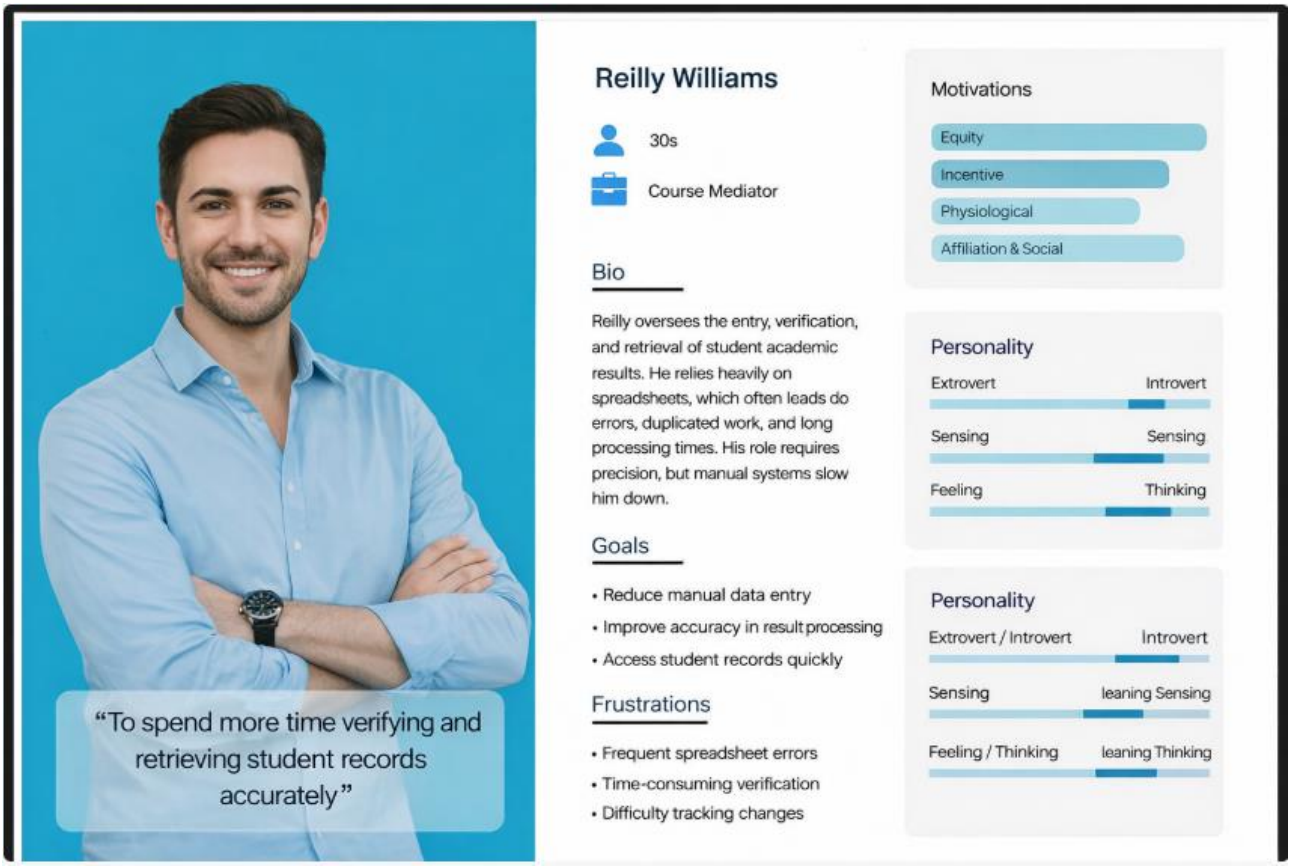
- Time-consuming verification
- Difficulty tracking changes

Motivations

- Equity
- Incentive
- Physiological
- Affiliation & Social

Arousal Personality

- Extrovert / Introvert — leaning Introvert
- Sensing / Intuition — leaning Sensing
- Feeling / Thinking — leaning Thinking



Core Modules of a Student Result Management System

User Authentication & Roles

Admin, Teacher, Student logic

Secure authentication (ID + password

Role-based access control.

Student Management

Add / update / delete student records

Student profile (name, roll no, class, department)

Student enrollments by academic year & semester

Course & Subject Management

Create courses and subjects

Assign subjects to classes/semesters

Credit hours and subject codes

Marks Entry & Evaluation

Teacher uploads marks (internal, external, practical)

Automatic total & grade calculation

Validation to prevent invalid entries

Result Processing

GPA / CGPA calculation

Pass/fail status

Ranking / merit list generation

Result Publication

Admin approval before publishing

Students view results online

Downloadable result sheets (PDF)

Reports & Analytics

Class-wise result analysis

Subject-wise performance reports

Student progress tracking

Notifications

Result published alerts

Email/SMS notifications (optional)

Exam & result announcements

Data Security & Backup

Secure database storage

Regular backups

Audit logs for changes

System Administration

Manage users and permissions

Academic year & semester setup

System configuration

Optional Advanced Features

Online re-evaluation request

Parent login portal

Graphical performance charts

Integration with attendance system

Problem Statement (How Might We)

How might we create a simple, reliable web application that allows educational staff to manage student results accurately and efficiently using a centralised database?

Initial Task Definition (Sprint 1)

The following tasks have been identified for Sprint 1:

- ❿ Form project team and agree on roles
- ❿ Create GitHub repository and add all members
- ❿ Set up task board for sprint tracking
- ❿ Define application idea and business context
- ❿ Define personas and problem statement
- ❿ Create Code of Conduct
- ❿ Prepare Sprint 1 proposal document

These tasks ensure the team is technically and organisationally ready to proceed.

Technology Overview

The application will be developed using:

- **Node.js** – backend runtime
- **Express.js** – client–server communication
- **PUG** – server-side templating
- **MySQL** – database storage
- **Git & GitHub** – version control
- **Docker** – deployment and environment consistency

Task Board

The team uses an online task board to manage sprint activities and track progress.

Task Board URL:

(Insert Trello / GitHub Projects link)

GitHub Repository

All source code and collaboration are managed through GitHub.

GitHub Repository URL:

(Insert GitHub link)

Code of Conduct

Our Pledge

We commit to maintaining a respectful, inclusive, and professional working environment throughout the project.

Expected Behaviour

- Attend sprint meetings and stand-ups
- Communicate respectfully and clearly
- Complete assigned tasks on time
- Support team members when challenges arise

Unacceptable Behaviour

- Harassment, discrimination, or offensive language
- Repeated failure to contribute
- Plagiarism or misuse of shared code

Conflict Resolution

- Issues will first be discussed within the team

- ❿ If unresolved, the issue will be escalated to the module tutor
- ❿ All members will have the opportunity to present their perspective

Conclusion

Sprint 1 establishes a strong foundation for the project by defining the business problem, application purpose, target users, and team workflow. The team is now prepared to move into Sprint 2 with clear objectives and a shared understanding of the project goals.

TASKBOARD LINK:

https://sharing.clickup.com/90152354961/b/h/2kyqzh4h-495/7fee7542143ce93

GITHUB LINK:

https://github.com/Hilary813/ManageSys